Paper II – Urology (MCh Urology, Summer 2017)

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II-1. Importance and management of Inguinal lymph nodes in CA penis.

Type: LAQ (25 marks)

Importance

- Presence and extent of inguinal metastasis is the **strongest prognostic factor** in penile SCC (squamous cell carcinoma).
- A long **locoregional phase** precedes distant spread, so the groin is a **curable station**; lymphadenectomy alone can cure.
- **Kroon (2005)**: early ILND (inguinal lymph node dissection) for impalpable nodes gave 3-year disease-specific survival about 84% vs about 35% when delayed until nodes were palpable (ENE (extranodal extension) about 20% vs 95%).
- Manage both groins **within 3 months** of diagnosis; aim for **at least 8 nodes per side**.
- About 80% 5-year survival with 2 or fewer unilateral nodes, no ENE, no pelvic nodes.

Lymphatic drainage

- **Superficial system** (prepuce, shaft skin) → superficial inguinal nodes. **Deep system** (glans, frenulum) → corona → within Buck's fascia → presymphyseal plexus → femoral triangle nodes.
- Presymphyseal crossover makes drainage **bilateral** and unpredictable in side; with one palpable groin, the other has micrometastasis in about 30%.
- **Stepwise, no skip**: superficial → deep inguinal → external iliac; so superficial status gates deep dissection and inguinal status gates pelvic dissection.
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**:

Zone	Related vessel
Central	Saphenofemoral junction
Superolateral	Superficial circumflex iliac vein
Inferolateral	Lateral femoral cutaneous and superficial circumflex veins
Superomedial (sentinel nodes mostly here)	Superficial external pudendal and superficial epigastric veins
Inferomedial	Great saphenous vein

- **Deep nodes** (1–3) medial to the femoral vein; **Cloquet's node** is the highest.

Staging (TNM (tumour–node–metastasis) 8th edition)

N	Clinical	Pathological
1	One mobile unilateral node	2 or fewer unilateral, no ENE
2	Mobile, 2 or more unilateral or bilateral	3 or more (unilateral or bilateral)
3	Fixed mass or pelvic nodes	ENE or pelvic nodes

- **T1a**: no LVI (lymphovascular invasion), no PNI (perineural invasion), not high grade. **T1b**: LVI or PNI or high grade.

Evaluation

- **Careful bilateral palpation**: cN0 or cN+ (number, side, mobile or fixed). Imaging (US (ultrasound), CT (computed tomography), MRI) only if examination is unreliable (obesity, previous groin surgery).
- **cN+ or high risk**: FDG-PET/CT (fluorodeoxyglucose positron emission tomography–CT) or CT chest–abdomen–pelvis.
- About half of palpable nodes are reactive, yet an antibiotic trial is **no longer recommended** (delays treatment). US-guided **FNAC (fine-needle aspiration cytology)** is specific; a negative result never ends the pathway.

Management of the impalpable groin (cN0): risk-based

Risk (EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023)	Primary	Groin
Low	pTis, pTa, pT1a G1	Surveillance
Intermediate	pT1a G2	Surveillance (compliant) or surgical staging
High	**pT1b or more**, G3	**DSNB (dynamic sentinel node biopsy)**; ILND if unavailable

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk, staged surgically (ILND first).
- **DSNB**: US ± FNAC first (a tumour-replaced node diverts flow → false negative); then 99mTc (technetium-99m) nanocolloid, blue dye and gamma probe; serial sectioning. False-negative about 7% at expert centres.
- **Modified ILND (Catalona)**: smaller field (nothing lateral to the femoral artery or below the fossa ovalis), **saphenous vein preserved**, no sartorius transposition; positive frozen section → complete ILND.
- **Also know:** superficial ILND; fascia-sparing ILND (FS-ILND); **VEIL (video-endoscopic inguinal lymphadenectomy)**/robotic: same yield, fewer wound complications, non-bulky nodes only; Cabanas anatomical sentinel biopsy abandoned.

Management of palpable nodes (cN+)

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Palpable node → US-guided FNAC
   positive → radical ILND same side + stage other side
              (superficial ILND, FS-ILND or DSNB)
   negative → excisional biopsy → negative: observe
Bulky (≥4 cm), fixed, bilateral or pelvic nodes
   → neoadjuvant TIP × 4 → responders: ilioinguinal resection
   → cisplatin-unfit: chemoradiation
```

- **Radical ILND** (cN2 or more; cN1 may have FS-ILND): incision 2 cm below and parallel to the inguinal ligament; flaps deep to **Camper's fascia**; boundaries inguinal ligament, sartorius, adductor longus; fascia lata opened; stay medial to the femoral artery (femoral nerve); **sartorius transposition** over the vessels; suction drains.
- **PLND (pelvic lymph node dissection)**: if **more than 2 inguinal nodes or ENE** (pN2–3); ipsilateral; external iliac commonest, then obturator.
- **Neoadjuvant TIP** (paclitaxel, ifosfamide, cisplatin), **Pagliaro (2010)**: objective response 50%; survival mainly in responders.
- **Adjuvant**: chemotherapy for pN2–3 (weak evidence); radiotherapy for multiple nodes or ENE (at least 50 Gy; 54 Gy for ENE).

Complications

- Wound infection, **flap necrosis**, dehiscence, seroma, **lymphocele**, **lymphoedema**, DVT/PE (deep vein thrombosis/pulmonary embolism).
- Reduced by transverse incision, saphenous sparing, drain removal by volume (<30 mL/day), thromboprophylaxis, minimally invasive access; sartorius transposition increases them.

Follow-up

- Examination 3-monthly for 2 years, then 6-monthly to 5 years (recurrence mostly within 2 years); then self-examination.

Diagram to draw: femoral triangle with Daseler's zones and Cloquet's node.

> **Key points to remember:**
> - Nodal status is the strongest prognostic factor; ILND can cure.
> - pT1b or higher: stage groins surgically (DSNB preferred) within 3 months.
> - Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
> - PLND if more than 2 nodes or ENE; bulky → neoadjuvant TIP (InPACT (International Penile Advanced Cancer Trial) testing the sequence).

II-2. Traumatic renal vascular parenchymal injuries : diagnosis and management.

Type: LAQ (25 marks)

Assumed meaning: diagnosis and management of traumatic renal parenchymal and renal vascular injuries.

Introduction

- Kidney is the **commonest urological organ injured** (up to 5% of trauma); **blunt 80–90%**.
- **Deceleration** stretches the hilum and UPJ (ureteropelvic junction): arterial intimal tear and thrombosis, vein disruption, pedicle avulsion.
- Main renal artery injury is rare (about 0.05% of blunt trauma).

Classification: AAST (American Association for the Surgery of Trauma) 2018

Grade	Injury
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration ≤1 cm, no extravasation
III	Laceration >1 cm without collecting-system injury; **any vascular injury or active bleeding contained within Gerota's fascia**
IV	Laceration into collecting system with extravasation; renal pelvis laceration or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's fascia; segmental or complete infarction (thrombosis)
V	**Main renal artery or vein laceration or hilar avulsion**; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for multiple injuries up to grade III. Change from 1989: main vessel injury moved from IV to V; pseudoaneurysm and AVF (arteriovenous fistula) now grade IV.

Clinical diagnosis

- Mechanism is the most important history. Signs: flank bruise, tenderness, **fractured ribs 9–12**, flank or low chest wound.
- **Haematuria** (first sample): gross, or microscopic (>5 RBC (red blood cells)/hpf (high-power field)) with shock. Degree does **not** match severity; it may be **absent in pedicle injury** and UPJ avulsion.

Indications for imaging (AUA (American Urological Association) and EAU (European Association of Urology))

1. Blunt trauma with gross haematuria.
2. Blunt trauma with microhaematuria and **SBP (systolic blood pressure) <90 mmHg** at any time.
3. Rapid deceleration (high-speed collision, fall), even without haematuria.
4. Penetrating trauma with likely renal injury, any haematuria.
5. Children: image liberally (hypotension appears late).

- Microhaematuria without shock or deceleration: observe without imaging.

Imaging

- **Contrast-enhanced CT (computed tomography)** is the gold standard. Arterial/nephrographic phase (~80 s) shows bleeding; **delayed excretory phase** (5–10

min) shows urinary extravasation.

- **CT red flags:**

Finding	Suggests
Medial haematoma	Pedicle (vascular) injury
Medial extravasation, no contrast in distal ureter	Renal pelvis or UPJ avulsion
Absent enhancement, cortical rim sign	Main renal artery occlusion
Active contrast blush	Arterial bleeding needing AE (angioembolisation)
Haematoma rim >4 cm, medial laceration	High risk of intervention

- **One-shot IVP (intravenous pyelography)** on table (2 mL/kg, film at 10 min) in the unstable patient: confirms a functioning contralateral kidney. FAST (focused assessment with sonography for trauma) is non-specific. Renal vein injury is poorly seen on CT.

Management

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Unstable despite resuscitation → laparotomy (or AE if team ready)
Stable → stage with CT
   Grade I–III → non-operative
   Grade IV–V  → non-operative in selected; AE for active bleeding
   Extravasation → observe; stent/nephrostomy if urinoma enlarges,
                   fever, or UPJ/pelvis injury
```

- **Non-operative**: bed rest until gross haematuria clears, serial haemoglobin; success >90% in blunt injury. Failure: instability, or falling haemoglobin despite 2–3 units of blood.
- **Angioembolisation**: superselective coils for active bleeding, pseudoaneurysm, AVF or delayed bleeding; repeat AE often succeeds.
- **Reimaging** (~48 h) for grade IV–V with deep laceration, or new fever, pain or falling haematocrit.

Indications for exploration

Absolute	Relative
Haemodynamic instability, no or transient response	Extravasation with devitalised parenchyma (>25%)
Expanding or pulsatile retroperitoneal haematoma	Arterial thrombosis
Suspected pedicle avulsion	Associated colon or pancreatic injury
UPJ avulsion	Incomplete staging

Operative technique

- Midline transabdominal; life-threatening injuries first.
- **Vascular control before opening Gerota's fascia**: lift small bowel; incise retroperitoneum over the aorta above the inferior mesenteric artery → left renal vein crossing the aorta → loop vein and both arteries; then reflect colon, open Gerota's fascia.
- **Renorrhaphy**: debride → suture-ligate vessels → watertight collecting-system closure → close defect over bolster → omental cover → drain. Polar injury: partial nephrectomy.
- **Nephrectomy** if unreconstructable or unstable. **Damage control**: pack, ICU (intensive care unit), relook at 24–48 h.

Renovascular injury

- **Main artery thrombosis** (intimal tear): unilateral with normal other kidney → observe (revascularisation rarely succeeds). **Solitary or bilateral → revascularise** (warm ischaemia ideally <120 min); endovascular stent for intimal flap.

- **Vein**: left renal vein can be ligated near the IVC (inferior vena cava) (gonadal, adrenal collaterals); right cannot. Repair with 5-0 vascular suture; segmental veins may be ligated.

Complications

- Delayed bleeding (within 3 weeks; AE), urinoma, abscess, **hypertension** (Goldblatt, **Page kidney**, AVF), hydronephrosis, loss of function.

Follow-up

- Blood pressure, urinalysis, creatinine; DMSA (dimercaptosuccinic acid) for grade III–V; imaging if new symptoms.

Diagram to draw: left renal vein crossing the aorta, with vessel loops.

> **Key points to remember:**
>
> - AAST 2018: segmental vessel injury grade IV; main vessel grade V.
> - Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
> - Stable patients, even grade IV–V: non-operative ± AE.
> - Revascularise only solitary or bilateral injuries.

II-3a. Irrigation fluids for TURP.

Type: SAQ (10 marks)

Ideal irrigant

- Transparent, **non-conducting** (monopolar), isotonic, non-toxic, non-haemolytic, cheap.
- **Why non-conducting?** In monopolar TURP (transurethral resection of the prostate), current flows loop → patient → pad; ionic saline would disperse it. **Bipolar** TURP completes the circuit in the instrument, so it uses **normal saline**.

Fluids

Fluid	Osmolality (approx.)	Points
Sterile water	0	Best vision, cheap; haemolysis, hyponatraemia; now avoided for TURP
Glycine 1.5%	~200 mOsm/kg	Commonest (UK, North America); glycine → ammonia, serine; transient blindness (retinal inhibitory neurotransmitter), encephalopathy
Sorbitol 3%	~165	Metabolised to fructose and glucose → hyperglycaemia; lactic acidosis
Mannitol 5%	~275	Near iso-osmolar; not metabolised; osmotic diuresis; volume overload in renal failure
Sorbitol 2.7% + mannitol 0.54%	~180	Continental Europe
Normal saline 0.9%	~308	Bipolar TURP, lasers (HoLEP, PVP); no TUR syndrome

- Plasma is about 290 mOsm/kg, so monopolar fluids are **hypo-osmolar**.

TUR (transurethral resection) syndrome

- **Mechanism**: prostatic venous pressure is ~10 mmHg; irrigant at higher pressure enters open sinuses → volume overload, **dilutional hyponatraemia** and hypo-osmolality → cerebral oedema; plus glycine toxicity.
- **Features**: confusion, nausea, vomiting, visual disturbance, hypertension then hypotension, bradycardia, seizures, coma.
- **Risk factors**: gland >45 mL, resection >90 minutes (VA (Veterans Affairs) study), high irrigation pressure, open sinuses or capsular perforation.
- **Prevention**: bag height ~60 cm (each extra 10 cm more than doubles absorption), continuous-flow resectoscope, time limit, spinal anaesthesia (awake patient), bipolar TURP. Ethanol 1% tagging can monitor absorption.
- **Treatment**: stop surgery, check sodium; Na <120 mEq/L serious. Furosemide for overload; **3% saline** 100 mL boluses (max 300 mL) for neurological signs; raise sodium by 4–6 mEq/L initially, not over 12 mEq/L in 24 h (risk of central pontine myelinolysis).

Bipolar advantage

- Meta-analyses: no TUR syndrome, less transfusion and clot retention; NNT (number needed to treat) about 50 to prevent one TUR syndrome.

Diagram to draw: monopolar vs bipolar current pathways.

Key points to remember:

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

II-3b. Pre operative investigations for renal donor.

Type: SAQ (10 marks)

Assumed meaning: evaluation of a living kidney donor.

Principles

- Donor safety first. **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** is **risk-based**: the donor's lifetime ESKD (end-stage kidney disease) risk is compared with the centre's threshold. Team includes an **independent donor advocate**.
- India: **THOA (Transplantation of Human Organs Act) 1994, amended 2011**: near relatives, or authorisation committee approval for others; swap donation allowed.

Sequence of evaluation

1. Compatibility

- ABO group, HLA (human leukocyte antigen) typing, **crossmatch** (CDC (complement-dependent cytotoxicity) and flow), DSA (donor-specific antibody); incompatible pairs → paired exchange.

2. Kidney function and urine

Test	Accept	Individualise	Decline
GFR (glomerular filtration rate), mL/min/1.73 m²	≥90	60–89	<60
Albuminuria (ACR (albumin–creatinine ratio), mg/g)	<30	30–100	>100

- GFR by two methods, one measured (DTPA (diethylenetriamine pentaacetic acid) or creatinine clearance).
- Urinalysis, culture; persistent microscopic haematuria → cystoscopy, imaging, ± kidney biopsy.

3. Medical and metabolic

- Blood pressure (ambulatory if borderline), fasting glucose, **HbA1c (glycated haemoglobin)**, OGTT (oral glucose tolerance test) if at risk; lipids, liver tests, BMI (body mass index).
- **Type 1 diabetes excludes**; prediabetes, controlled hypertension and obesity: individual risk.
- **Stones**: metabolic work-up; bilateral or recurrent usually exclude.

4. Infection screen

- HIV (human immunodeficiency virus), HBsAg (hepatitis B surface antigen), anti-HBc, HCV (hepatitis C virus), CMV (cytomegalovirus), syphilis; **tuberculosis** screen.

5. Cardiopulmonary: ECG (electrocardiogram), chest X-ray, echo; stress test if indicated.

6. Cancer screening by age: Pap smear, mammogram, PSA (prostate-specific antigen), stool test or colonoscopy.

7. Anatomy

- **CT (computed tomography) angiography**: number of vessels, early branching, stones, masses.
- **Split function** (DTPA or DMSA (dimercaptosuccinic acid)) if asymmetric: the **better kidney stays with the donor**; left preferred (longer vein).

8. Psychosocial: voluntary, no coercion or payment.

Counselling figures (Muzaale 2014)

- Lifetime ESKD 90 per 10,000 donors vs 14 in matched non-donors; mortality 3 per 10,000.

Also know: APOL1 (apolipoprotein L1) testing in African ancestry; healthy older (marginal) donors acceptable.

Diagram to draw: donor work-up flowchart.

Key points to remember:

- KDIGO 2017: risk-based; decline GFR <60 or ACR >100.
- ABO and crossmatch first; CT angiography for anatomy.
- Leave the better kidney with the donor.
- THOA: near relatives or authorisation committee.

II-3c. UW solution.

Type: SAQ (10 marks)

Introduction

- **University of Wisconsin solution** (Belzer, late 1980s): the **gold standard** cold-storage solution for kidney, liver and pancreas.

Principles of cold preservation

- Cooling to **4 °C** cuts metabolism to roughly a tenth, but does not stop it.
- Ischaemia → ATP (adenosine triphosphate) loss → **Na^+/K^+-ATPase pump failure** → cell swelling; anaerobic glycolysis → acidosis; on reperfusion, **oxygen free radicals** (xanthine oxidase) → ischaemia–reperfusion injury, seen clinically as DGF (delayed graft function).
- A solution must therefore prevent swelling and oedema, buffer acid, scavenge radicals and supply ATP precursors.

Composition (per litre)

Component	Amount	Purpose
Lactobionate (K^+ salt)	100 mmol	Impermeant anion; prevents cell swelling; chelates calcium
Raffinose	30 mmol	Impermeant trisaccharide
Hydroxyethyl starch	50 g	Colloid; prevents interstitial oedema
KH_2PO_4	25 mmol	Buffer
$MgSO_4$	5 mmol	Membrane stability
Adenosine	5 mmol	ATP precursor
Glutathione	3 mmol	Antioxidant
Allopurinol	1 mmol	Xanthine oxidase inhibitor; fewer free radicals
Insulin, dexamethasone, antibiotic	Small	Additives

- **Intracellular-type**: K^+ about 125 mmol/L, Na^+ about 30 mmol/L; osmolality ~320 mOsm/kg; pH 7.4.

Drawbacks

- **Viscous** (starch): slow flush. **High K^+**: flush before reperfusion (hyperkalaemia, vasospasm). May crystallise (filter).

Comparison

Feature	UW	HTK (Custodiol)	Euro-Collins
Type	Intracellular (high K^+)	Low K^+, low Na^+	Intracellular
Impermeant	Lactobionate, raffinose	Mannitol	Glucose (enters cells; weaker)
Buffer	Phosphate	**Histidine** (strong)	Phosphate
Colloid	HES (hydroxyethyl starch)	None (low viscosity)	None
Others	Antioxidants, adenosine	Tryptophan, ketoglutarate	—

- Kidney outcomes with UW and HTK (histidine–tryptophan–ketoglutarate) are similar; Euro-Collins largely abandoned. **Also know:** Celsior, IGL-1, Marshall's citrate.

Practical use

- Living donor, very short cold ischaemia: cold lactated Ringer's suffices; deceased donor or exchange: UW or HTK.
- Keep cold ischaemia short (ideally <24 h): DGF and thrombosis rise with time.
- **Hypothermic machine perfusion** (Belzer MPS (machine perfusion solution), gluconate-based) reduces DGF vs static storage (**Moers 2009**) without long-term survival gain; normothermic perfusion experimental.

Diagram to draw: ischaemia cascade with the UW component acting at each step.

> **Key points to remember:**
>
> - Lactobionate and raffinose (impermeants), HES (colloid).
> - Allopurinol, glutathione (antioxidants); adenosine (ATP).
> - High K^+: flush before reperfusion.
> - HTK: histidine buffer, low viscosity, low K^+.

II-3d. Uro Dynamic Studies.

Type: SAQ (10 marks)

Definition and principles

- UDS (urodynamic studies) test bladder and urethral function in storage and voiding.
- ICS (International Continence Society): **decide the question first**, **reproduce the symptom**; a normal study does not exclude disease.
- Before invasive UDS: symptom score, 3-day **frequency–volume chart**, free uroflow and PVR (post-void residual); treat UTI (urinary tract infection) first.

Components

Test	Method	Main values
Uroflowmetry	Voided volume ≥150 mL	Bell curve normal; Qmax <10 mL/s (men) suggests obstruction or weak detrusor; plateau = stricture
PVR	Bladder scan or catheter	Emptying
Filling cystometry	Bladder and rectal lines, fill with saline 30–100 mL/min (adults); **Pdet = Pves − Pabd**; zero at atmosphere, transducer at symphysis	Sensation, capacity, **DO (detrusor overactivity)**, **compliance** = ΔV/ΔPdet (<20 mL/cmH_2O impaired)
Leak point pressures	ALPP (abdominal LPP): cough or Valsalva at 150 mL	Low ALPP = weak sphincter (ISD (intrinsic sphincter deficiency))
	DLPP (detrusor LPP)	**>40 cmH_2O** risks upper tracts (McGuire)
Pressure–flow study	Void with lines in	High pressure–low flow = obstruction; low–low = underactivity
EMG (electromyography)	Perineal surface electrodes	DESD (detrusor–external sphincter dyssynergia)
UPP (urethral pressure profile)	Catheter withdrawn	Limited value
Videourodynamics	Above with contrast and fluoroscopy	VUR (vesicoureteric reflux), bladder neck, site of obstruction

Indices (men)

- **BOOI (bladder outlet obstruction index)** = PdetQmax − 2Qmax: **>40 obstructed**, 20–40 equivocal, <20 unobstructed.
- **BCI (bladder contractility index)** = PdetQmax + 5Qmax: **>150 strong**, 100–150 normal, **<100 weak**.
- Women: no validated nomogram (Nitti video criteria).

Indications

- Male LUTS (lower urinary tract symptoms) before surgery when unclear (young men, equivocal flow, failed TURP (transurethral resection of the prostate)); complex, mixed or recurrent incontinence.
- **Neurogenic bladder** (spinal injury, MS (multiple sclerosis), spina bifida), radiation cystitis, before augmentation.

- Children: neurogenic bladder, PUV (posterior urethral valves); fill at **5–10% of expected capacity per minute**; expected capacity = (age + 2) × 30 mL.

Urodynamic risk factors

- Impaired compliance, DESD, DISD (detrusor–internal sphincter dyssynergia), high-pressure DO throughout filling.

Also know: ambulatory UDS; bethanechol test; antibiotic prophylaxis for risk groups.

Diagram to draw: normal trace: Pves, Pabd, Pdet, flow, EMG.

> **Key points to remember:**
> - Pdet = Pves − Pabd.
> - BOOI >40 obstructed; BCI <100 weak.
> - DLPP >40 cmH_2O: upper-tract risk.
> - Ask a question first; reproduce the symptom.

II-3e. Azospermia

Type: SAQ (10 marks)

Assumed meaning: azoospermia.

Definition

- **No sperm in the ejaculate** after **centrifugation** of the sample, on **at least two** semen analyses; 10–15% of infertile men.

Classification

Type	Cause	Clues
Pre-testicular (hypogonadotropic)	Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone or anabolic steroids**	Low FSH (follicle-stimulating hormone), LH (luteinising hormone), testosterone
Testicular (NOA (non-obstructive azoospermia), spermatogenic failure)	Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy or radiation, idiopathic	Small soft testes, **FSH >7.6 IU/L**, long axis ≤4.6 cm
Post-testicular (OA (obstructive azoospermia))	CBAVD (congenital bilateral absence of the vas deferens), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction)	Normal testes, **FSH ≤7.6**, dilated epididymis
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Evaluation

- History (mumps, surgery, drugs, testosterone); examination (testis size, **vasa**, epididymis, varicocele).
- **Hormones**: FSH (reflects germ-cell mass), LH, testosterone, prolactin.

```
Azoospermia → vasa absent? → CBAVD → CFTR testing (both partners), renal US
            → volume <1 mL? → sperm in post-ejaculate urine → retrograde
                            → none → TRUS for EDO
            → normal volume: FSH ≤7.6 + testis >4.6 cm → obstructive
                             FSH >7.6 + small testes → spermatogenic failure
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- **Genetics** (AUA/ASRM (American Society for Reproductive Medicine) guideline): **karyotype** and **Y microdeletion** in NOA; Klinefelter (47,XXY) commonest. **AZF (azoospermia factor) a or b deletion: no sperm, do not operate**; AZFc: sperm often found. **CFTR (cystic fibrosis transmembrane conductance regulator)** in CBAVD.
- **TRUS (transrectal ultrasound)**: seminal vesicle >1.5 cm, midline cyst → EDO.
- **Testis biopsy**: combined with sperm extraction and freezing.

Management

- **Hypogonadotropic**: stop testosterone; **hCG (human chorionic gonadotropin) ± FSH**; dopamine agonist for prolactinoma.
- **Obstructive**: **vasovasostomy** or **vasoepididymostomy** (microsurgical); **TURED (transurethral resection of ejaculatory ducts)** for EDO; or sperm retrieval (MESA (microsurgical epididymal sperm aspiration), PESA, TESA) + ICSI (intracytoplasmic sperm injection).

- **Non-obstructive**: **micro-TESE (microdissection testicular sperm extraction)** + ICSI; sperm in about half. Repair clinical varicocele; hormonal optimisation (off-label).
- Retrograde ejaculation: sympathomimetics, or sperm from alkalinised urine.
- No sperm: donor sperm or adoption; genetic counselling.

Diagram to draw: hypothalamic–pituitary–testis axis with the three levels of failure.

> **Key points to remember:**
> - Confirm on two centrifuged samples.
> - FSH ≤7.6 and testis >4.6 cm → obstructive.
> - Karyotype and Y microdeletion in NOA; AZFa/b: do not retrieve.
> - NOA: micro-TESE + ICSI.

II-3f. Radiotherapy for CA prostate.

Type: SAQ (10 marks)

Radiobiology

- Prostate cancer has a **low α/β ratio (~1.5 Gy)**, lower than the rectum, so larger fractions favour tumour kill over late toxicity: the basis of **hypofractionation**.

EBRT (external beam radiotherapy)

- **IMRT (intensity-modulated radiotherapy)**/VMAT (volumetric modulated arc therapy) with **IGRT (image-guided radiotherapy)** (fiducials, cone-beam CT (computed tomography)) is standard, allowing dose escalation with less toxicity.

Schedule	Dose per fraction	Example	Trial
Conventional	1.8–2 Gy	78 Gy in 39	—
Moderate hypofractionation (standard)	2.25–4 Gy	**60 Gy in 20**	**CHHiP**: non-inferior to 74 Gy in 37
SBRT (stereotactic body RT)	≥5 Gy	36.25 Gy in 5	**PACE-B**, HYPO-RT-PC

- **Also know:** focal MRI boost (FLAME); protons; **hydrogel rectal spacer** reduces rectal toxicity.

Brachytherapy

- **LDR (low dose rate)**: permanent ^{125}I (140–160 Gy) or ^{103}Pd seeds via TRUS (transrectal ultrasound)-guided perineal template; monotherapy for low and favourable intermediate risk.
- **HDR (high dose rate)**: temporary ^{192}Ir afterloading (e.g. 27 Gy in 2).
- **Boost** (EBRT 45 Gy + implant) for unfavourable intermediate or high risk: **ASCENDE-RT**: LDR boost halved biochemical failure, with more urinary toxicity.
- **Contraindications**: IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s, recent TURP (transurethral resection of the prostate), pubic arch interference, prior pelvic RT (radiotherapy), inflammatory bowel disease.

ADT (androgen deprivation therapy) with RT, by risk

Risk	ADT
Low, favourable intermediate	None
Unfavourable intermediate	**4–6 months**
High / locally advanced	**24–36 months** (EORTC 22863); abiraterone if very high risk (STAMPEDE)

- RT added to ADT improves survival in locally advanced disease (**SPCG-7, NCIC PR.3/MRC PR07**).

Post-prostatectomy RT

- **Adjuvant RT** (pT3, positive margin) improves control (EORTC 22911, ARO 96-02) and survival (SWOG 8794).
- **RADICALS-RT, RAVES, GETUG-AFU 17** (ARTISTIC meta-analysis): **early salvage at PSA (prostate-specific antigen) 0.1–0.2 ng/mL** equals adjuvant RT and spares most men RT.
- Salvage 64–66 Gy + ADT (**RTOG 9601**: 2 years bicalutamide improved survival) ± pelvic nodes (SPPORT).

Outcomes

- **ProtecT** (2016; 15-year 2023): equally low prostate-cancer mortality with RT, surgery or active monitoring; RT more bowel effects, surgery more incontinence and ED (erectile dysfunction).
- Failure after RT: **Phoenix definition** = nadir + 2 ng/mL.

Toxicity

- Acute: frequency, dysuria, diarrhoea. Late: **proctitis** with bleeding (6–18 months; sucralfate, argon plasma coagulation), cystitis, stricture, ED, second cancers.

Diagram to draw: TRUS-guided perineal template brachytherapy.

> **Key points to remember:**
>
> - α/β ~1.5 Gy: basis of hypofractionation (CHHiP 60 Gy in 20).
> - ADT: 4–6 months unfavourable intermediate, 24–36 months high risk.
> - Early salvage (PSA 0.1–0.2) = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
> - Phoenix: nadir + 2 ng/mL.

Rapid Revision Sheet – Paper II

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

II-1. Importance and management of Inguinal lymph nodes in CA penis. (LAQ)

- Nodal status is the strongest prognostic factor; ILND can cure.
- pT1b or higher: stage groins surgically (DSNB preferred) within 3 months.
- Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
- PLND if more than 2 nodes or ENE; bulky → neoadjuvant TIP (InPACT (International Penile Advanced Cancer Trial) testing the sequence).

II-2. Traumatic renal vascular parenchymal injuries : diagnosis and management. (LAQ)

- AAST 2018: segmental vessel injury grade IV; main vessel grade V.
- Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
- Stable patients, even grade IV–V: non-operative ± AE.
- Revascularise only solitary or bilateral injuries.

II-3a. Irrigation fluids for TURP. (SAQ)

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

II-3b. Pre operative investigations for renal donor. (SAQ)

- KDIGO 2017: risk-based; decline GFR <60 or ACR >100.
- ABO and crossmatch first; CT angiography for anatomy.
- Leave the better kidney with the donor.
- THOA: near relatives or authorisation committee.

II-3c. UW solution. (SAQ)

- Lactobionate and raffinose (impermeants), HES (colloid).
- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High K^+: flush before reperfusion.
- HTK: histidine buffer, low viscosity, low K^+.

II-3d. Uro Dynamic Studies. (SAQ)

- Pdet = Pves − Pabd.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 cmH_2O: upper-tract risk.
- Ask a question first; reproduce the symptom.

II-3e. Azospermia (SAQ)

- Confirm on two centrifuged samples.
- FSH ≤7.6 and testis >4.6 cm → obstructive.
- Karyotype and Y microdeletion in NOA; AZFa/b: do not retrieve.
- NOA: micro-TESE + ICSI.

II-3f. Radiotherapy for CA prostate. (SAQ)

- α/β ~1.5 Gy: basis of hypofractionation (CHHiP 60 Gy in 20).
- ADT: 4–6 months unfavourable intermediate, 24–36 months high risk.

- Early salvage (PSA 0.1–0.2) = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- Phoenix: nadir + 2 ng/mL.